

Deep learning analysis of smart meter data from a small sample of room air conditioners facilitates routine assessment of their operational efficiency

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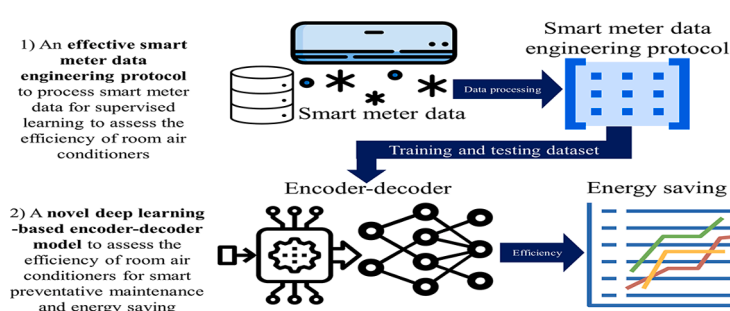
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HIGHLIGHTS

- Limited RAC samples are available to learn from smart meter data about efficiency.
- We applied zero filter & resampling to treat smart meter data and increase samples.
- Deep learning models were developed to evaluate RAC efficiency.
- Our protocol and model can accurately assess RAC operational efficiency.
- Energy can be saved through smart preventative maintenance based on our solution.

GRAPHICAL ABSTRACT



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ABSTRACT

Room air conditioners (RACs) are crucial household appliances that consume substantial amounts of electricity. Their efficiency tends to deteriorate over time, resulting in unnecessary energy wastage. Smart meters have become popular to monitor electricity use of home appliances, offering underexplored opportunities to evaluate RAC operational efficiency. Traditional supervised data-driven analysis methods necessitate a large sample size of RACs and their efficiency, which can be challenging to acquire. Additionally, the prevalence of zero values when RACs are off can skew training. To overcome these challenges, we assembled a dataset comprising a limited number of window-type RACs with measured operational efficiencies from 2021. We devised an intuitive zero filter and resampling protocol to process smart meter data and increase training samples. A deep learning-based encoder-decoder model was developed to evaluate RAC efficiency. Our findings suggest that our protocol and model accurately classify and regress RAC operational efficiency. We verified the usefulness of our approach by evaluating the RACs replaced in 2022 using 2022 smart meter data. Our case study demonstrates that repairing or replacing an inefficient RAC can save electricity by up to 17 %. Overall, our study offers a potential energy conservation solution by leveraging smart meters for regularly assessing RAC operational efficiency and facilitating smart preventative maintenance.

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1. Introduction

Space conditioning (heating and cooling) typically accounts for ~50 % of total building energy use [1]. In 2021, the demand for space cooling specifically accounted for ~16 % of the total electricity consumption in buildings. In the past twenty years, the demand for space cooling has been growing at an annual rate of 4 %, twice as quickly as that for lighting and water heating [2]. With more frequent heatwaves induced by climate change, the need for air conditioners will undoubtedly increase [3–5]. Room air conditioners (RACs) are a popular choice for residential homes due to their affordability, ease of installation, and effectiveness in cooling small spaces. As air cooling becomes increasingly prevalent, RACs will continue to account for a large share of home energy consumption, making their efficiency management key to saving energy and reducing greenhouse gas emissions [6–8].

Unfortunately, the significance of RAC energy efficiency during the operational phase is often overlooked by both manufacturers and consumers, leading to late maintenance and thus substantial energy waste [9]. On the one hand, while energy efficiency labels offer informed, sustainable choices to buyers of RACs, they provide little information on energy efficiency when the machines are actually in use [10]. On the other hand, individuals typically make decisions to clean, maintain, or replace their RAC based on their personal thermal comfort, which may potentially cause a delay in upgrading or repairing inefficient machines, resulting in energy waste. Therefore, it is desirable to develop measures for tracking RAC energy efficiency during the operational phase to offer users information about the RAC conditions and facilitate in-time actions to save energy [11].

The increasingly popular smart meters for home appliances offer a promising opportunity as they can monitor electricity consumption in real time and provide timely feedback to consumers [12–14]. Moreover, the vast amount of data collected by smart meters fosters great potential for data-driven analysis of energy use at home [15–17]. To this end, various studies have explored data-driven approaches, leveraging machine learning algorithms, to analyze smart meter data for electricity prediction and RAC management, including diagnosing abnormal

electricity use of AC [18,19] and controlling RAC operation [20,21]. Using the RAC smart meter data in Sydney, Australia, the contribution of RAC to home's peak demand was quantified to evaluate the potential peak demand reduction through demand response [22]. However, the added value of smart meters in evaluating the RAC energy efficiency and supporting maintenance measures remains significantly underdeveloped [23]. Setting up in-field measurements of RAC operation and performance with sensors to monitor the units inside a RAC can be time-intensive and costly [9]. It is challenging for researchers and practitioners to gather a substantial number of RAC samples with operational data on outdoor and indoor units as well as electricity use for operational efficiency assessment [24]. Consequently, due to these data limitations, the method of using smart meter data to assess RAC efficiency is left unexplored.

In this regard, this study aims at utilizing the smart meter data to evaluate RAC energy efficiency and inform smart preventive maintenance, for which Fig. 1 provides an overview of challenges and solutions. There are three challenges in sample preparation to learn from smart meter data to evaluate RAC energy efficiency (Fig. 1a). The first challenge is that for supervised data-driven analysis to evaluate RAC efficiency, a large dataset that includes RAC operation efficiency, smart meter data, and other essential feature data is typically necessary [25]. However, it has been a challenge for RAC manufacturers, utilities, property managers, and consumers to obtain massive samples of RACs and their operation efficiency (Fig. 1a) [26,27]. The second challenge is that since RACs only operate occasionally and in specific seasons, the collected data contains many zero values, generating significant noise in our dataset; they provide no valuable information for data-driven models to comprehend the efficiency of RAC. The third challenge is that although the dataset encompasses a considerable length of time, the number of RACs with valid records was limited [28]. These caveats impose challenges in applying a supervised data-driven model to evaluate RAC efficiency with smart meter data.

In this study, we proposed an intuitive zero filter and resampling protocol to formulate the sample based on smart meter data and expand the samples for training (Fig. 1b). We constructed a dataset of ninety-six

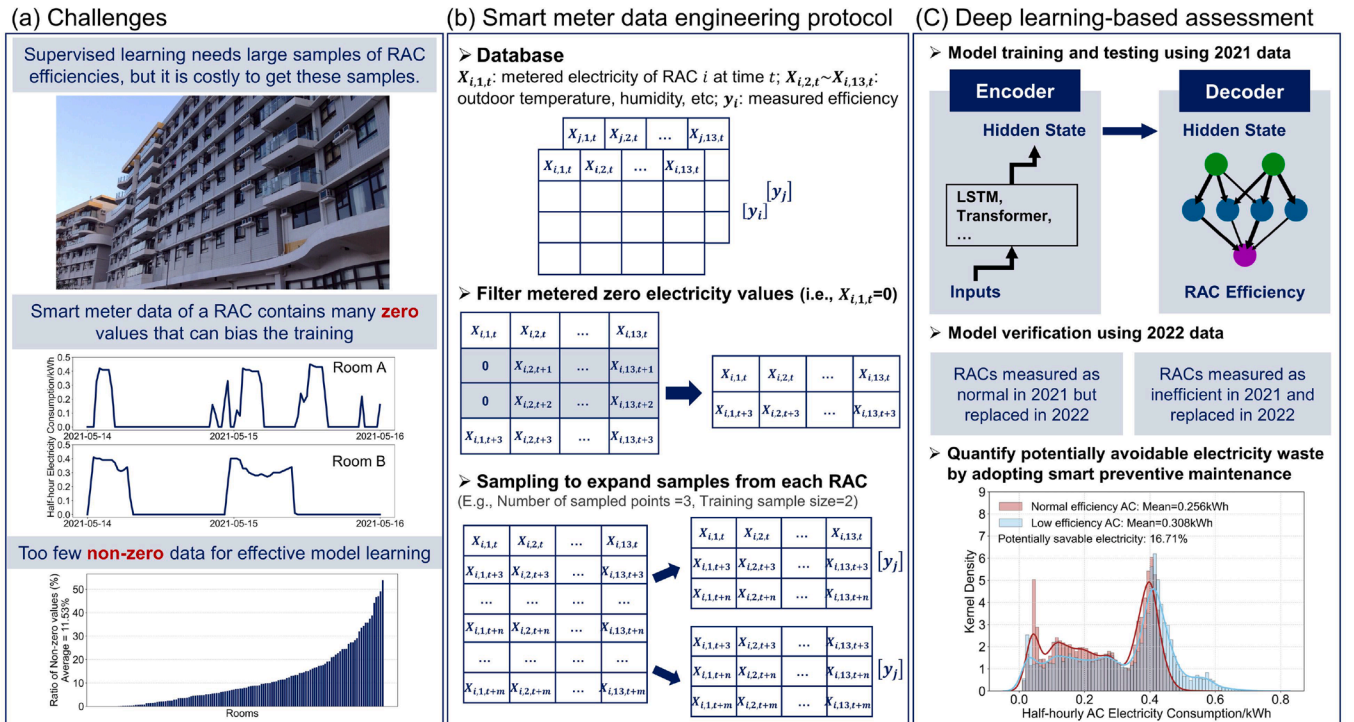


Fig. 1. Research overview includes (a) the challenges of using smart meter data to evaluate the efficiency of RACs; (b) smart meter data engineering protocol we propose; (c) deep learning-based assessment model development, verification, and benefits.

window type RACs in a university hall from December 2020 to August 2021. We measured the efficiency in the summer of 2021 while the rooms were unoccupied. We recorded the indoor temperature and humidity before and after turning on RAC and the time it took for the room temperature and humidity to become stable. We calculated the efficiency as a ratio of heat removed to power consumption. The protocol initiates by eliminating zero values. Then from a single RAC, a sub-set of non-zero data points is selected and chronologically to formulate a new sample. The number of sampled points determines how many non-zero values will be sampled from a RAC. This sampling procedure is repeated for different RACs to increase samples, based on the training sample size. The effects of the number of samples points and training sample size on the model's performance are evaluated accordingly.

Subsequently, an encoder–decoder model was developed using deep learning architectures to classify and regress the RAC operational efficiency (Fig. 1c). To evaluate the accuracy of our models, we designed two settings. Setting I splits the samples from all RACs to train and test models. This means that all RACs contribute samples for both training and testing. Setting II splits RACs and the samples from a portion of the RACs are used to train models and the samples from the remaining RACs are used to evaluate the trained models. Setting I assesses whether a model accurately evaluates the efficiency of a RAC given a new sample from the same RAC, while Setting II checks whether the model performs well when measuring the efficiency of a new RAC. We evaluated the model's performance in both classification and regression tasks. In the classification task, the RACs were labeled as "low efficiency" if their measured efficiency was below the mean value of all RACs. On the other hand, if the measured efficiency was equal to or above the mean, the RACs were considered "normal." In the regression, the model predicts the measured efficiency. Our analytical model demonstrates extraordinary performance in both settings for RAC efficiency classification and regression tasks, with more reliable results generated under Setting I than Setting II. We further verified our model accuracy against the RAC replacement records and quantified the potentially avoidable energy waste using our smart meter data engineering protocol and deep learning model.

Overall, this study contributes to an innovative solution to assess the energy efficiency of RACs throughout their operational phase with our smart meter data engineering protocol and deep learning-based encoder–decoder model. Our study provides an important contribution to overcoming the limitations of smart meter data to prepare and feed samples to deep-learning algorithms to evaluate appliances' energy efficiency. Our smart meter data engineering protocol can be expanded to use smart meter data to evaluate the operational efficiency of other appliances at home or factory, which requires more validations in future studies. Finally, our study advances our understanding of the added value of smart meter data in saving energy and promoting sustainability [29].

The rest of our paper consists of four more sections. In Section 2, we describe our research design, data collection, training task definition, smart meter data treatment and sample expansion, deep learning model construction, training and evaluation, verification of model assessment, and the analysis of potentially avoidable energy waste if RACs with low efficiency can be repaired or replaced in time. Results are reported in Section 3 including the impact of the number of sampled points and training sample size on model's performance, the accuracy of different deep learning architectures, verification, and the potentially avoidable energy waste. Discussions are presented in Section 4 about future improvement and application of our smart meter data engineering protocol and deep learning model in smart preventive maintenance followed by the conclusion in Section 5.

2. Methods

2.1. Data collection

We collected three types of data inputs from December 30, 2020, to August 15, 2021. These data include: (1) half-hourly electricity consumption smart meter data for non-inverter window-type RACs; (2) half-hourly meteorological data (i.e., outdoor temperature, precipitation, relative humidity, and irradiance) from the Atmospheric & Environmental Real-time Database; and (3) anonymized Wi-Fi connection counts in rooms to approximate the number of occupants. Table 1 summarizes the input features we created based on these three types of data inputs for deep learning model training.

To collect RAC efficiency as the training output, we conducted the field measurements in the summer of 2021 when the university dorm

Table 1

Definitions and statistics of input features extracted from our collected dataset.

Input features	Explanation	Minimum	Maximum	Mean	Std
<i>E_{AC}</i>	Half-hourly electricity consumption (kWh)	0.010	0.720	0.298	3.676
<i>Temperature</i>	Half-hourly outdoor temperature (°C)	5.431	36.223	26.243	3.676
<i>Irradiance</i>	Hourly outdoor irradiance (W/m ²)	0.011	808.269	84.996	165.594
<i>Precipitation</i>	Hourly precipitation (mm)	0.000	1.092	0.004	0.034
<i>Humidity</i>	Hourly outdoor relative humidity (%)	0.702	99.852	84.047	12.104
<i>WIFI</i>	Average hourly Wi-Fi connection count to approximate the number of people in the room	0.000	14.000	2.351	1.467
<i>Prev_one</i>	Total AC electricity consumption of the RAC in the previous 30 min	0.000	0.720	0.277	0.165
<i>Prev_three</i>	Total AC electricity consumption of the RAC in the previous 90 min	0.010	2.120	0.822	0.424
<i>Prev_five</i>	Total AC electricity consumption of the RAC in the previous 150 min	0.010	3.430	1.267	0.706
<i>Prev_one_on</i>	Whether the RAC was turned on in the previous 30 min (True/ False)	0.000	1.000	0.889	0.315
<i>Prev_two_on</i>	Whether the RAC was turned on in the previous one hour (True/ False)	0.000	1.000	0.790	0.407
<i>Next_one_on</i>	Whether the RAC was turned on in the next 30 min (True/ False)	0.000	1.000	0.889	0.315
<i>Next_two_on</i>	Whether the RAC was turned on in the next one hour (True/ False)	0.000	1.000	0.790	0.407

rooms were not occupied. Fig. 2 shows the typical indoor setting of rooms. A compact refrigerator is the only major appliance in the room, and it was unplugged for all rooms. We first recorded outdoor and indoor temperature and humidity when we entered the room using the Q-Trak air quality monitor [30]. The accuracy of the monitor is $\pm 0.5^\circ\text{C}$ and $\pm 3\%$ for temperature and relative humidity, respectively. The response time is 30 s. We turned on the RAC and recorded the time when the indoor temperature and humidity stabilized. The RACs in the dorm were non-inverter window-type air conditioners, which are commonly found in Hong Kong households. Popular brands include GREE, York, and Panasonic. We recorded the rated power of RACs. We calculated the RAC efficiency as a ratio of heat removed to power consumption (Eq. (1) and Supplementary Information RAC Efficiency Calculation).

$$\eta = \frac{Q(\Delta T) + Q(\Delta H)}{W \times t} \quad (1)$$

Where, η is the measured efficiency; Q is the heat removed as a function of room temperature change (ΔT) and humidity change (ΔH); W is the rated power of RAC; and t is the consumed time until the room temperature and humidity stabilize. We rescaled the measured efficiency between 0 and 1 for model training and testing (Supplementary Fig. S1 for the distribution).

Our efficiency measurement is relatively simple and straightforward as compared to the rigorous setting of measuring the operations of subunits in the RAC. However, our measurement is feasible in time that can allow us to run the measurements and assess the efficiency of 96 RACs in a university hall within a week. In addition, according to the replacement records in 2022, 8 out of 12 replaced RACs were rated inefficiency by us. This consistency between our assessment and the replacement needs suggest that our efficiency measurement is reliable to distinguish the efficiency among RACs. This allows us to use the measured efficiency as the training output.

2.2. Smart meter data engineering protocol for sample expansion

Our smart meter data engineering protocol starts by removing zero-value data points. In our study, this step eliminated 94 % of the data, resulting in 72,463 non-zero data points. Each data point is represented as a 13-dimensional vector, denoted by $x = [x_1, x_2, x_3, \dots, x_{13}]$, and associated with the RAC efficiency, denoted by y . From the non-zero value data points of one RAC, we randomly sampled a subset of data points and rearranged the selected data points in chronological order to create a matrix of data points, as shown in Eq. (2). Each matrix served as one data sample for model training. The number of data points in each sample (i.e., the number of sampled points) is a hyperparameter to be tuned to create samples of varied sizes for training. We set another parameter to control the total number of data samples from RACs (i.e., training sample size). The number of samples drawn from each RAC is proportional to its number of non-zero data points. Through this treatment and sampling approach, we extended the total number of samples available for training deep learning models to assess the RAC efficiency. Each sample contains denser information of electricity consumption and

holds temporal integrity to train deep learning models.

$$x_{\text{sample}} = \begin{bmatrix} x_1^1 & \dots & x_{13}^1 \\ \vdots & \ddots & \vdots \\ x_1^{\text{number of sampled points}} & \dots & x_{13}^{\text{number of sampled points}} \end{bmatrix}, y_{\text{sample}} = y \quad (2)$$

2.3. Deep learning model construction

We then construct deep learning models that learn from these transformed data samples to assess RAC efficiency. Specifically, we adopted an encoder-decoder deep learning structure, in which the encoder extracts the embeddings from data samples, and the decoder performs downstream tasks of classification and regression. Five popular deep learning architectures were utilized as the backbone model for the encoder, including long short-term memory (LSTM) [31], bidirectional long short-term memory (BiLSTM) [31], transformer [32], LSTM-transformer, and BiLSTM-transformer. A fully connected neural network was used as the decoder to perform classification or regression tasks. This design offers several advantages. Firstly, it allows for the possibility of replacing the encoder with a more advanced architecture, thereby enhancing the overall performance of the model. Additionally, by training a separate decoder to accommodate different objectives instead of training the entire model from scratch, our choice maximizes flexibility. The unsupervised RAC efficiency clustering algorithm proposed by Wang et al. [33] is used as the baseline for comparison against our model. Details of the baseline model and our deep learning model configurations are shown in Supplementary Information Model Architectures. We optimized the deep learning models with fine-tuned hyperparameters to maximize accuracy (Supplementary Table S2). In addition to hyperparameters for each encoder model, the impacts of the number of sampled points and training sample size on classification and regression performance were investigated (Supplementary Table S2). Increasing the sample size and number of training samples can theoretically lead to a more accurate model by increasing data density and quantity. We proposed classification and regression tasks of evaluating RAC efficiency. In the classification task, the RACs were labeled as "low efficiency" if their measured efficiency was below the mean value of all RACs. On the other hand, if the measured efficiency was equal to or above the mean, the RACs were considered "normal." Deep learning based binary classifiers were developed to distinguish whether the RAC is normal or inefficient. In the regression task, deep learning models were developed to estimate the RAC measured efficiency after rescaling.

2.4. Model training and evaluation settings

Following model constructions, we train our models on the pre-transformed data samples and manually measured efficiencies to let them accurately identify and predict RAC efficiencies. We designed two settings for model training and validation. In the first setting, we utilized 80 % of non-zero smart meter data points from all RACs to generate training data samples, while the remaining 20 % from all RACs was used to create validation and testing samples (Setting I). This approach



Fig. 2. University hall room indoor setting in our case study: the floor area is the same for 2 and 3 beds.

ensured no overlaps in smart meter data to generate the training, validation, and testing samples through our proposed smart meter data treatment. In other words, the model is trained on all RACs and then assessed on unseen data samples from these RACs. Thus, Setting I is to evaluate the performance of our model when evaluating new samples from previously seen RACs. In the second setting, we randomly selected 80 % RACs to produce training samples, while the remaining 20 % RACs were reserved for validation and testing (Setting II). To evaluate the impact of sample size and number of training data, the RACs were randomly sampled for each test. In this setting, we evaluated the performance of our models to evaluate the efficiency of an unseen RAC, which indicates the generalizability of our smart meter data engineering protocol and deep learning model. Setting II simulates a zero-shot evaluation setting that evaluates data samples whose source RACs were not presented in training [34]. The models were trained using a meticulously crafted setup and extensive computational resources (Supplementary Information Experiment Setup). For classification tasks, we reported classification accuracy and F_1 score, while for regression tasks, we used root mean square error (RMSE) and R^2 score.

2.5. Verification of inefficient RAC detection in 2022

We used our models trained with 2021 data to evaluate the efficiency of RACs replaced in 2022 to verify our model for smart preventive maintenance. The RACs that were replaced in 2022 were included in our model training under Setting I. We consider these RACs to be inefficient, which is why they were replaced. If our model can accurately identify their inefficiencies using the 2022 data, it demonstrates that our deep learning model can provide preventive maintenance suggestions over time. The verification is to quantify the percentage of these RACs accurately assessed based on 2022 smart meter data. To validate this, we utilized our top-performing classifier from Setting I to classify the efficiency of these replaced RACs in 2022 using their smart meter data from that year, prior to replacement. We drew one hundred data samples for each replaced RAC, and the efficiency was determined by the majority of these one hundred evaluations. To provide effective preventive maintenance suggestions, our deep learning model must possess the capability to (1) detect inefficiencies in RAC performance over time and (2) identify a decline in RAC efficiency over time. Thus, we calculated the percentage of accurate assessment for RACs identified as inefficient in both 2021 and 2022, and those that transitioned from being efficient in 2021 to inefficient in 2022, respectively. The verification helps determine the effectiveness of our model in evaluating both RAC inefficiency and RAC efficiency decrease, which are essential for smart preventive maintenance.

2.6. Potentially avoidable energy waste

To assess the effectiveness of our smart meter data engineering protocol and deep learning model, we conducted an evaluation to quantify the potential energy savings achieved through smart preventive maintenance using our deep learning model in 2022. Our primary goal was to apply our protocol and model to the 2022 data, identifying RACs with inefficiencies and measuring the additional energy consumed by these units compared to normal ones. This analysis allowed us to determine the extent to which our protocol and model contribute to energy savings. We used our best efficiency classifier, which was trained under Setting I on 2021 data, to classify the efficiency of 127 RACs in the university hall based on their smart meter data in 2022. For RACs replaced in 2022, only data prior to their replacements were used. These RACs were separated into two classes, normal and inefficient, based on the evaluation. We compared the average half-hourly electricity consumption of the two classes to compute the potentially avoidable energy waste and the percentage of energy that can be saved for the four periods (0:00AM–8:00AM, 8:00AM–14:00PM, 14:00PM–18:00PM, and

18:00PM–24:00PM). The calculation aims to verify the usefulness of our proposed protocol and model for assessing RAC efficiency based on smart meter data to save energy.

3. Results

3.1. Model accuracy in assessing RAC efficiency

In Setting I, 80 % of non-zero data points from all RACs were selected to produce the training samples, and the remaining 20 % were for generating validation and testing samples. The number of samples drawn from each RAC was proportionate to its number of non-zero smart meter data points. The heat maps in Fig. 3 display four average metric values of the five deep learning-based encoder–decoder models (*Methods Deep Learning Model Construction*) in classifying and regressing RAC's efficiency. Model performance improves as the number of sampled points and the training sample size increases. Our models achieve nearly 100 % accuracy in classifying RAC's efficiency, and R^2 of 0.98 in regressing RAC's efficiency. These results suggest that our model is dependable to evaluate the efficiency of RAC using a new sample from the same RAC. If the RAC is normal or inefficient, the model evaluation will reflect its normal or inefficient operational with the new sample from this RAC.

Fig. 4 presents the models' performances in Setting II. In this setting, we divided the RACs into two groups: 80 % of the RACs, along with their corresponding samples, were selected for training the models. The remaining 20 % of the RACs, along with their samples, were reserved for testing and validation purposes. In both classification and regression tasks, the performances across different numbers of sampled points and training sample sizes exhibit stochastic fluctuations. For the classification task, a deepening color gradient from the upper left to the lower right suggests that a larger number of sampled points and an increased training data size led to improved performance. The F_1 score heatmap contains grids with a value of 0, which does not necessarily imply poor model performance. When randomly splitting all the RACs to construct the training, validation, and test sets, the test samples can contain either positive or negative samples exclusively, causing the F_1 score to approach zero. In the regression task, the RMSE decreases as the training sample size increases. However, the R^2 score heatmap reveals that most values are close to zero, with a maximum R^2 score of only 0.62. These findings indicate the models' deficient performance in estimating the efficiency value of unseen RACs. Results in Setting II suggest that our smart meter data engineering protocol and deep learning-based encoder–decoder models can be used to classify the efficiency of unseen RACs of the same type. However, it remains a challenge to accurately compute the efficiency values of unseen RACs of the same type [34].

3.2. Model performance comparison

Wang et al. [33] proposed an eXtreme Gradient Boosting (XGBoost) model to predict hourly electricity consumption of RACs and used the Shapley value of outdoor temperature (i.e., the impact of outdoor temperature on electricity consumption estimate) to distinguish the inefficient RACs from normal ones. We compared the performance of our models with the XGBoost baseline in the classification task. Table 2 summarizes the average performance metrics of the five deep learning models across different numbers of sampled points and training sample sizes. Our proposed models significantly outperformed the XGBoost baseline in the classification task and offers a new innovative solution to use smart meter data to assess the efficiency of RACs.

Fig. 5 illustrates the performance of the five deep learning models with different encoders in Setting I. As the number of sampled points and training sample size increases, the performance of the models uniformly improves. In instances where the number of sampled points and training sample size are small, both the LSTM and BiLSTM models outperform

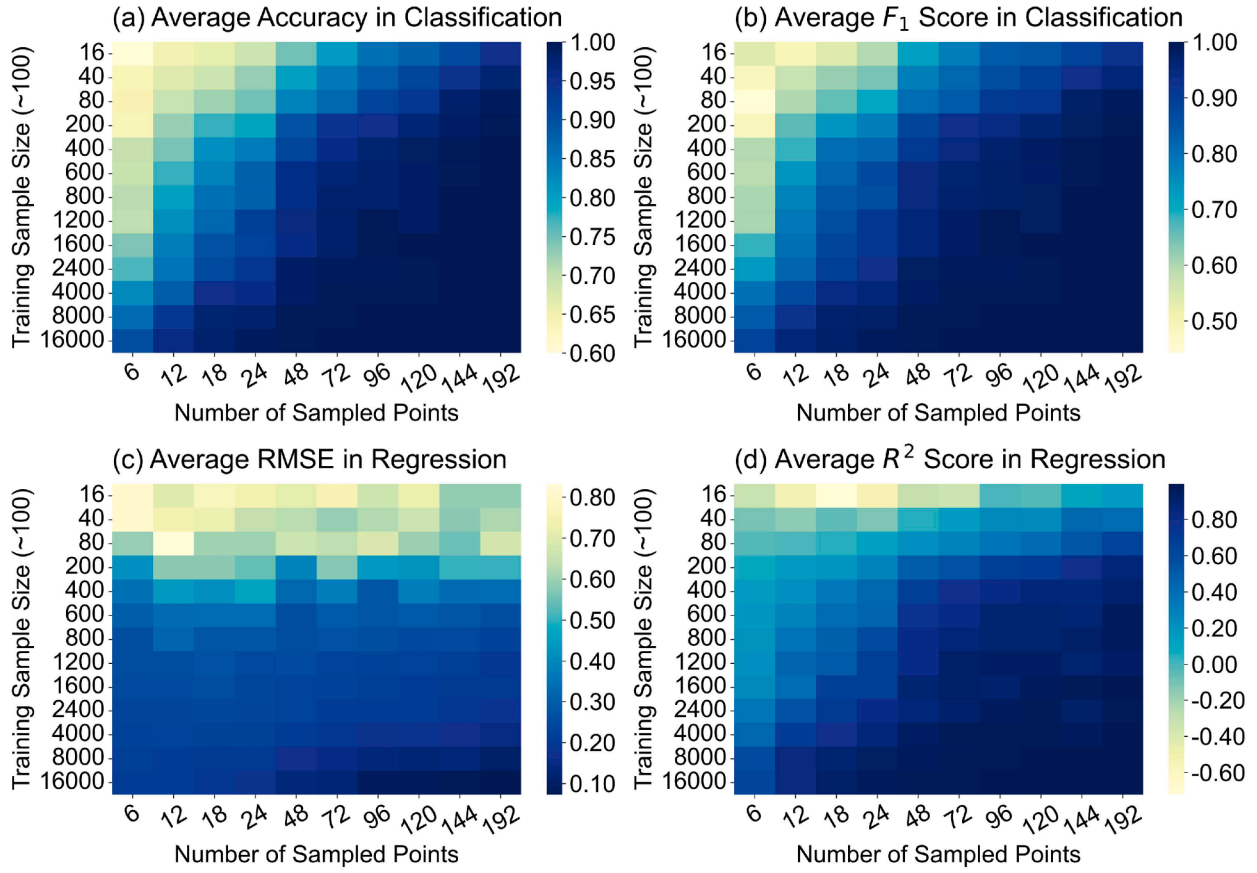


Fig. 3. Four metric statistics for the average model performance in setting I classification and regression tasks, including accuracy, F_1 score, RMSE, and R^2 score.

transformer-based models in both classification and regression tasks. This observation suggests that recurrent neural network (RNN) structures, including LSTM and BiLSTM, learn more efficiently on data samples with low information density (i.e., the number of sampled points is small). As the number of sampled points and training sample size increases to larger values, transformer-based models become as competitive as LSTM and BiLSTM models.

Fig. 6 illustrates the performance of the five deep learning models in Setting II. In contrast to Fig. 5, there is no consistent improvement as the number of sampled points and training sample size increases. In Setting II, the transformer-based models outperform LSTM and BiLSTM models, particularly when combined with LSTM or BiLSTM layers. The attention mechanism in transformer blocks enhances the model's generalizability and analytical ability, making LSTM-Transformer or BiLSTM-Transformer the preferred model encoders to classify the efficiency of unseen RACs. All models perform better under Setting I than Setting II, suggesting that it is more dependable to include RACs and their historical smart meter data in training to regularly evaluate these RACs for smart preventive maintenance. While the model remains possible to evaluate the efficiency of unseen RACs, classifying the efficiency is more useful and dependable than estimating the efficiency values.

3.3. Verification of inefficient RAC detection in 2022

The training-testing results using 2021 data highlight that if an RAC is normal and has been included in the training, our model in Setting I can correctly evaluate its normal status using a new sample from this RAC as long as its efficiency remains unchanged. The same applies to the inefficient RAC. We want to verify this implication using 2022 data. But most importantly, we want to verify whether our model can evaluate the transition from normal to inefficient status from smart meter data as the RAC becomes less efficient.

We applied our best model in Setting I and evaluated the efficiency of nine RACs that had been replaced in 2022, and we assume that these RACs replaced in 2022 are considered inefficient. We measured these RACs in 2021, and our models in Setting I included these RACs in training. The verification is to quantify the percentage of these RACs accurately assessed based on 2022 smart meter data. Out of five RACs we measured as inefficient in both 2021 and 2022, our model using 2022 half-hourly meter data correctly identified three as inefficient (Fig. 7a). The error in one of the two incorrect evaluations can be attributed to the limited meter data. This confirms our conclusion that if the RAC's efficiency remains unchanged, our model in Setting I can provide consistent assessments.

Out of four RACs we measured as normal in 2021 but becoming inefficient in 2022, our model using 2022 half-hourly meter data classified three as inefficient. This verification suggests that our model can evaluate the transition from normal to inefficient status (Fig. 7b). If an RAC becomes inefficient, our model can identify this transition based on the samples from smart meter data from this RAC. This implies that our models in setting I can be applied for preventive maintenance. In the beginning, smart meter data and measured RAC efficiency are collected to train the model we proposed. The model can help regularly evaluate the efficiency of RAC based on smart meter data and inform users to repair or replace the inefficient RAC on time. The model should be updated once RACs are repaired or replaced to maintain a high accuracy, since our results of Setting II highlights our model is not as accurate in evaluating new RACs. In general, our study offers a promising solution of using smart meter data to support preventive maintenance of RACs.

3.4. Potentially avoidable electricity waste

The potentially avoidable electricity waste from upgrading the

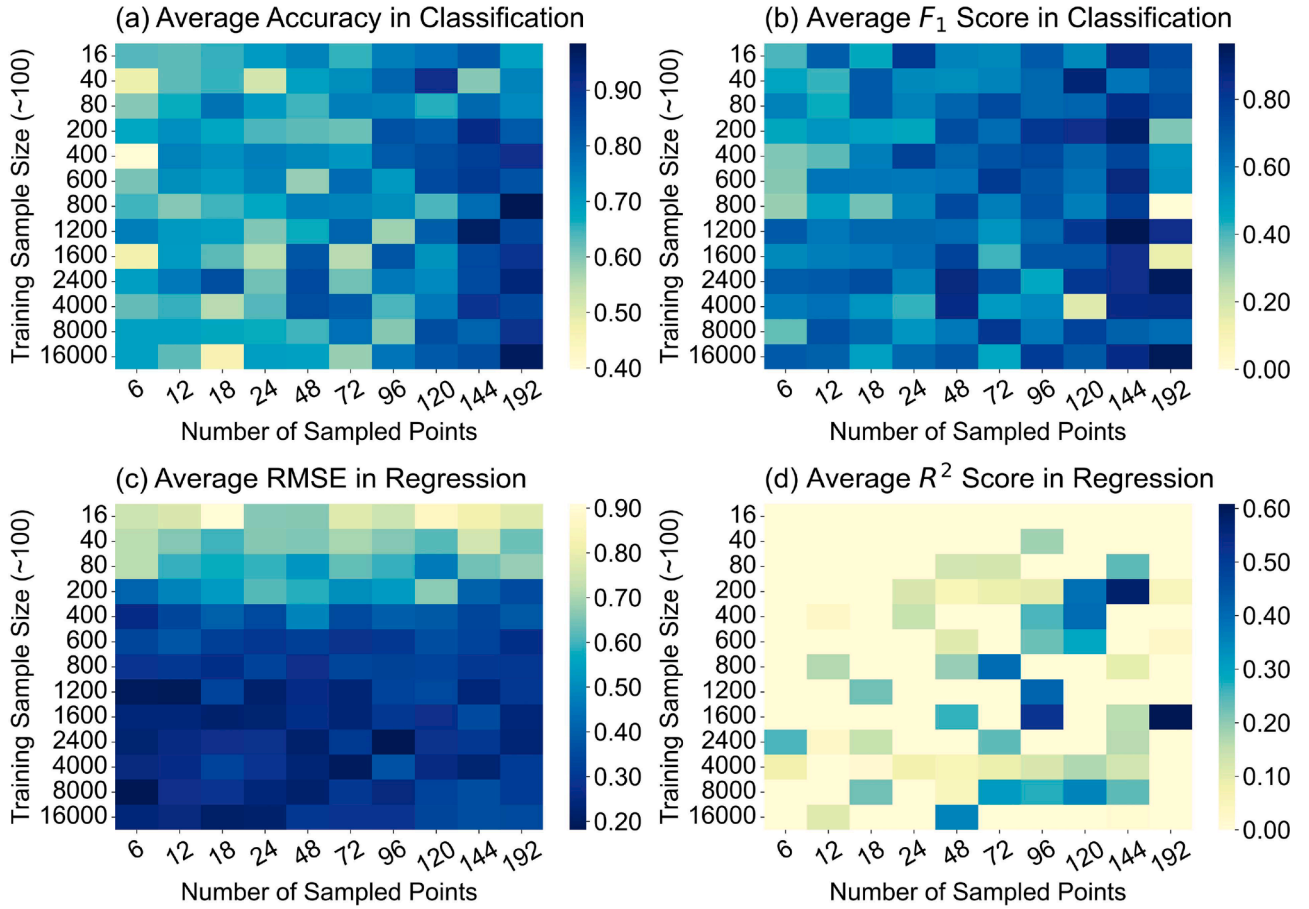


Fig. 4. Four metric statistics for the average performance of deep learning-based encoder-decoder models in setting II classification and regression tasks, including accuracy, F_1 score, RMSE, and R^2 score.

Table 2

Performances of XGBoost baseline versus our deep learning models for the RAC efficiency classification task.

Model	Accuracy		F_1 score	
XGBoost	0.5824		0.5766	
LSTM	0.9177	Setting I	0.8988	Setting I
BiLSTM	0.9229	Setting II	0.9046	Setting II
Transformer	0.8858		0.8586	
LSTM-Transformer	0.8792		0.8541	
BiLSTM-Transformer	0.8833		0.8533	

inefficient RAC on time was quantified in our case study to suggest the value of our proposed protocol and model for energy efficiency and sustainability. We used the best model under Setting I to evaluate the efficiency of all RACs based on smart meter data in 2022. Fig. 8 shows the kernel density distribution of half-hourly electricity consumption of the RACs with normal and low efficiency in four time periods of a day. The potentially avoidable electricity use of inefficient RACs is up to 21.07 % during the period of 00:00AM–8:00AM, and the saving is at least 9.43 % in other time periods, which corresponds to 0.028–0.067 kWh of electricity per half-hour. Overall, 16.72 % of electricity can be saved by identifying and fixing inefficient RACs on time, suggesting the sustainability value of applying our protocol and model to support preventive maintenance.

Normal RACs consume more electricity in the afternoon and evening when the outdoor temperature is high. The electricity use of normal RACs tends to be lower during the late night and morning when the outdoor temperature is relatively cooler. A decrease in outdoor

temperature leads to a decrease in indoor cooling demand and a higher coefficient of performance. Consequently, the electricity consumption of normal RACs decreases accordingly. In contrast, the electricity use of inefficient RACs is relatively constant regardless the influence of outdoor temperature. The patterns of electricity use of normal and inefficient RAC provide another verification of our smart meter data engineering protocol and deep learning model in evaluating the operational efficiency of RACs.

4. Discussion

4.1. The value of our smart meter data engineering protocol and deep learning model for assessing RAC's operational efficiency

Our study proposed a protocol for processing smart meter data from RACs and expanding training samples based on a limited number of RACs. Leveraging our protocol, the deep learning-based encoder-decoder model demonstrates superior performance for the classification and regression of evaluating the efficiency of RACs. This is a significant improvement over conventional supervised learning that is often impeded by the shortage of RACs records and their operational efficiency. Our protocol retains useful information by dropping many zero values in the smart meter record and increasing the sample size for each RAC by randomly sampling the non-zero electricity use data points. The high accuracy in Setting I suggests that electricity use of RACs can contain useful information to evaluate operational efficiency. Putting more records of electricity use in a sample can be more useful to judge the operation of RACs under various outdoor and indoor environments. That is why the increase in the number of sampled points contributes to

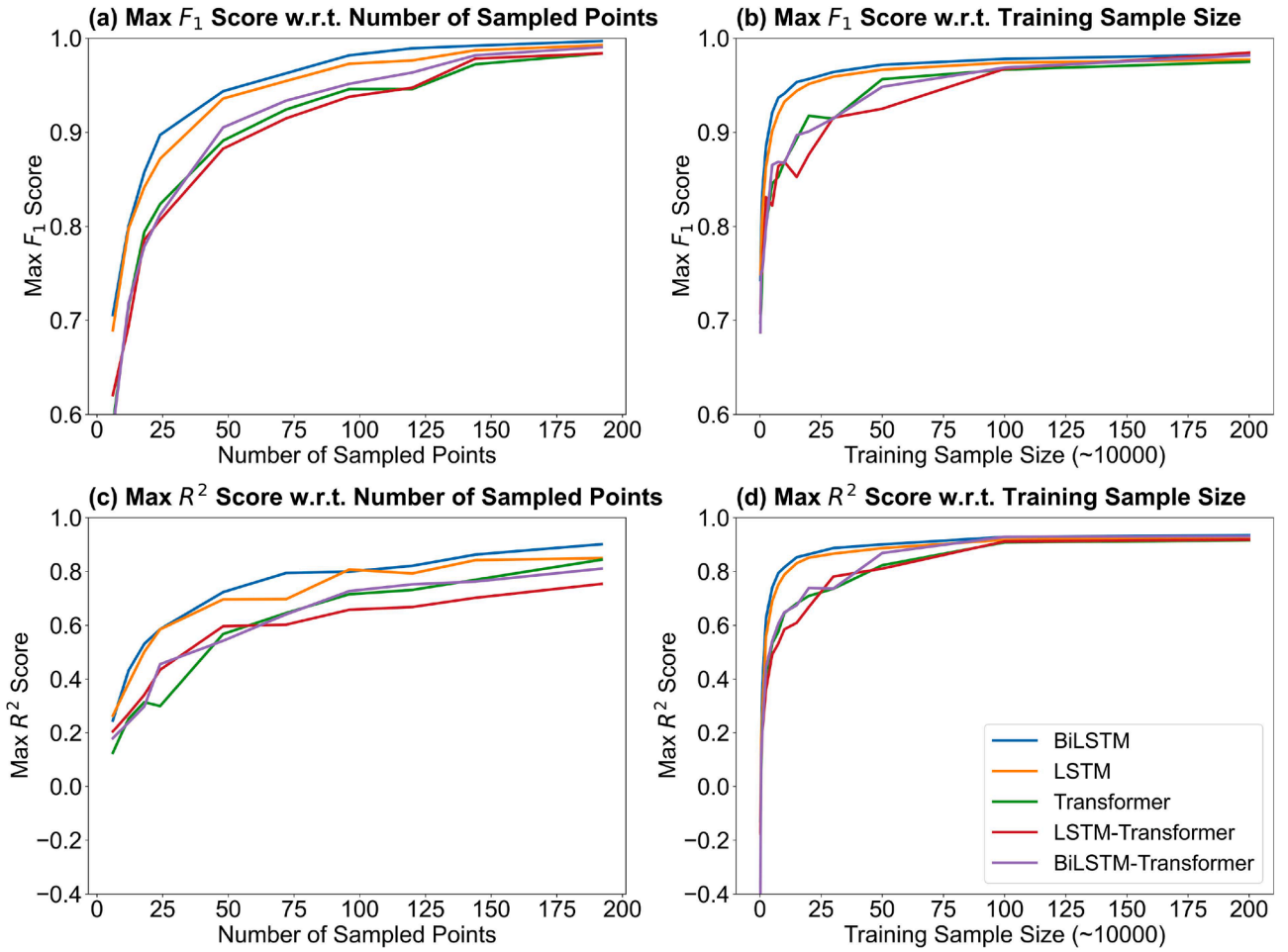


Fig. 5. Five models' maximum F_1 and R^2 scores on setting I classification and regression tasks with respect to different numbers of sampled points and training sample sizes.

better accuracy in both classification and regression in Setting I.

Our deep learning-based encoder-decoder model offers an alternative solution that reduces the complexity and saves labors and cost over traditional field measurements for efficiency assessment. This is particularly beneficial for large buildings with numerous RACs, streamlining data collection and efficiency assessment. Out of our two settings for formulating the training and testing samples, Setting I demonstrates more reliability than Setting II. This result highlights the usefulness of using smart meter data for regular assessment and preventive maintenance [35]. Together with the cost of replacing a RAC, the replacement decision can be properly supported as well through regular assessment and energy saving quantification on a case-by-case basis.

4.2. The application of the protocol and model for routine assessment

To initialize this application in real life, a set of RACs and their operational efficiency should be collected, which can be performed by both property managers and RAC manufacturers. Based on our findings, a small number of RACs' records and measures of their operational efficiency would be sufficient. Next, a deep-learning-based encoder-decoder model can be trained under Setting I to regularly evaluate the existing RACs. Property managers of university halls and rental apartments can use this trained model for smart preventive maintenance. Also, RAC manufacturers can provide a service built on this model to users to learn their RACs' operational efficiency. The records of RAC maintenance and replacement received by property managers and RAC manufacturers can be used as the sample of inefficient RACs to

update and improve the model accuracy. Repaired and new RACs should be added to model updates with the assumption that the new ones are always normal. These RACs are considered as unseen samples, and therefore, the model needs to be updated to ensure accurate predictions, suggested by the results of Setting II. This approach can progressively increase the samples of RACs and their measured efficiency to update the model, for smart preventive maintenance. The accuracy of evaluating the operational efficiency of unseen RACs will also be improved consistently as the less satisfactory performance in Setting II is due to the limited number of RACs in this study. Eventually, there is no need to include the newly installed RAC to update the training as required by Setting I. To achieve this industrial-scale application, our study creates a solution to initialize the effort of using smart meter data for smart preventive maintenance [36].

In Setting I, our models exhibited strong performance in both the classification and regression tasks. In the classification task, we labeled the efficiency of RACs by comparing their measured efficiency with the average efficiency of all the RACs in the dataset. However, it is important to note that in practical applications, there may be established technical standards or criteria to determine the efficiency level based on precise measurements. These standards can provide a more objective and standardized approach to classifying the efficiency of RACs. On the other hand, the regression task, which directly estimates the efficiency values, can be particularly informative for decision-making processes. By predicting the actual efficiency levels, decision-makers can gain valuable insights into the extent of the efficiency drop in RACs. This information can guide decisions on maintenance, repair, or replacement

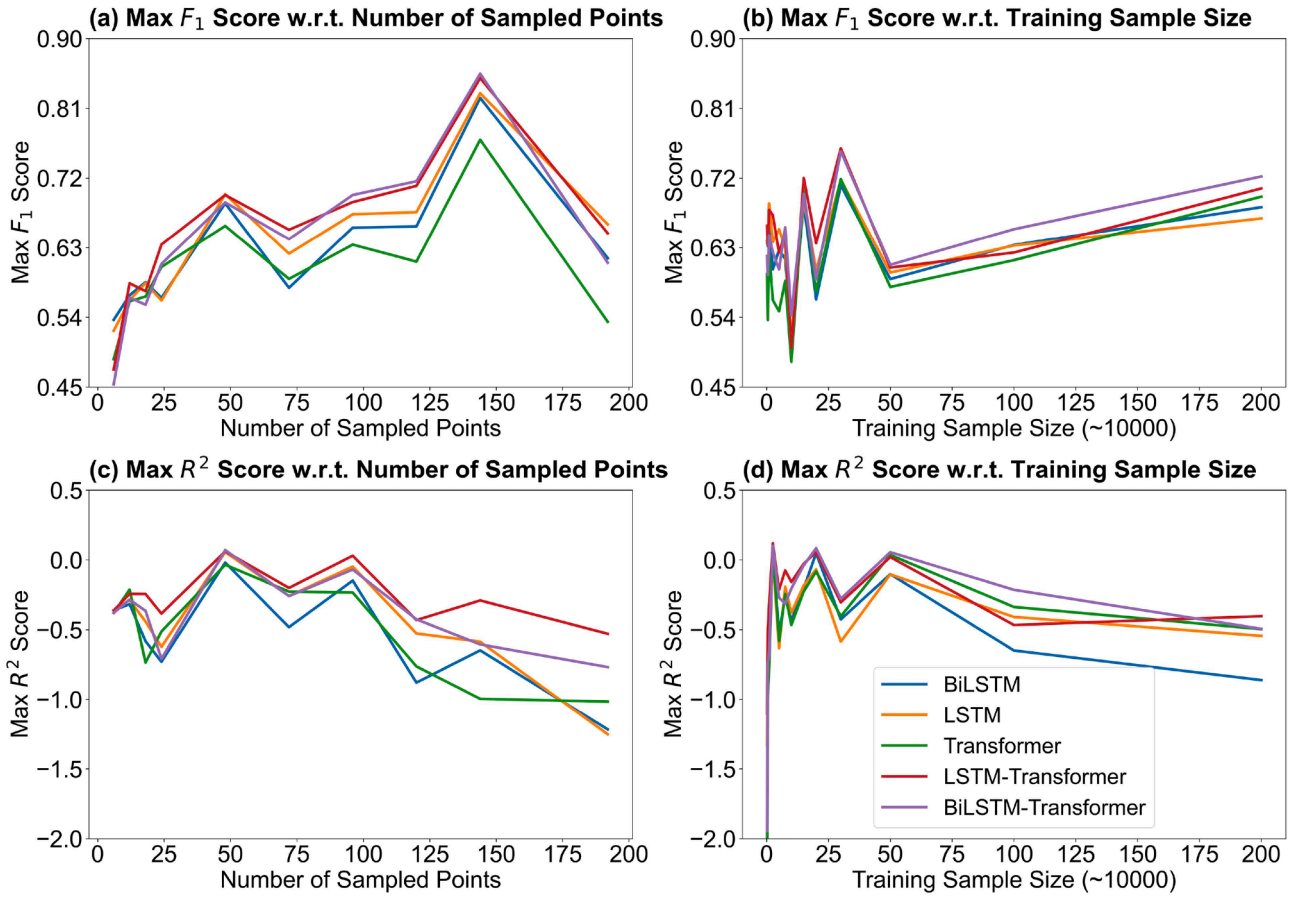


Fig. 6. Five models' maximum F_1 and R^2 score on setting II classification and regression tasks with respect to different numbers of sampled points and training sample sizes.

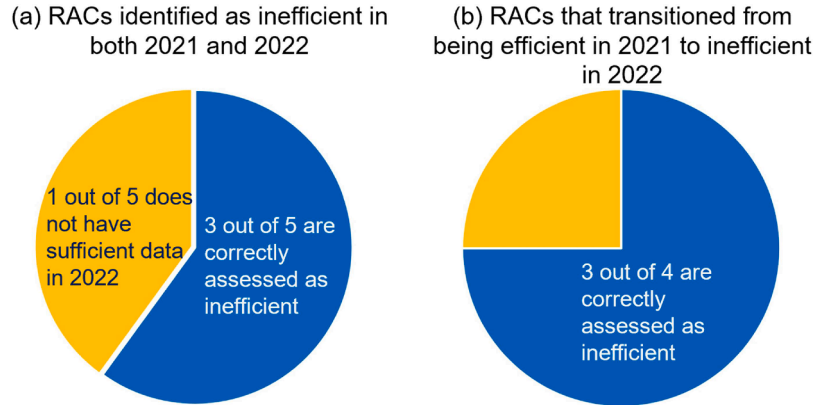


Fig. 7. Percentage of accurate assessment based on 2022 smart meter data: (a) RACs identified as inefficient in both 2021 and 2022, and (b) RACs that transitioned from being efficient in 2021 to inefficient in 2022.

of RAC units based on the observed efficiency levels.

4.3. Future needs for improvement

First, more case studies are needed to verify the applicability of our protocol in processing smart meter data and increasing the training samples based on a limited number of appliances to evaluate operational efficiency. In our study, we only investigate one type of RAC, and future studies can explore other RACs (e.g., split type) [37]. Our verification using 2022 smart meter data has a small number of RACs and more rigorous verification is needed in the future. Second, additional

potential input features including power disturbances and occupancy behavior, have the potential to improve model performance, particularly in regression tasks. The challenge is to find effective monitoring solutions to collect these input features. Nevertheless, in future applications where such additional features become accessible, they can be seamlessly integrated into our developed protocol and model, following a similar workflow encompassing model identification, model training, and model verification. Last, while our efficiency measurement is considered efficient and reliable, it is necessary to conduct more rigorous measurements to confirm the most effective measurement solution. With more accurate measurements as the training outputs, this

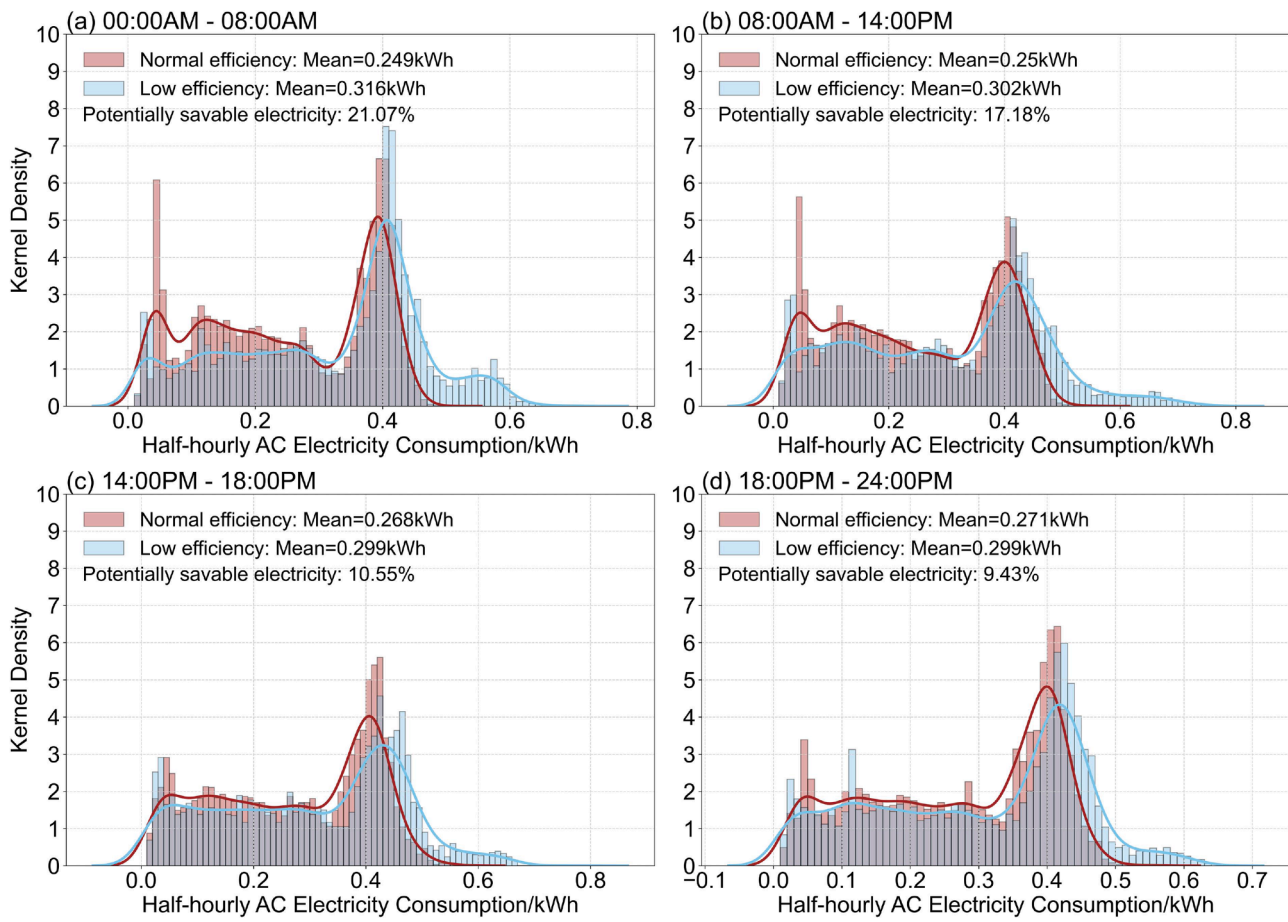


Fig. 8. Kernel density distribution of half-hourly electricity consumption of RACs with normal and low efficiency during four time periods of the day, including 00:00AM–08:00AM, 08:00AM–14:00PM, 14:00PM–18:00PM, 18:00PM–24:00PM, in 2022.

will substantially improve the reliability of our smart meter data engineering protocol and deep learning model for regular assessment and preventive maintenance.

5. Conclusion

In this study, we developed an innovative solution consisting of a smart meter data engineering protocol and deep learning models that can use smart meter data from a limited sample of RACs for assessing the operational efficiency. Our study contributes to the mining of added value from smart meters for energy conservation and improved sustainability. Our computational solution can lead to a potential service for property managers, RAC manufacturers, and utility companies to leverage smart meter data to improve the awareness of operational efficiency of RACs for smart preventive maintenance and energy saving. Furthermore, our smart meter data engineering protocol can inspire similar trials to use smart meter data when only limited samples of appliances, equipment or services are available for training and evaluation. Moreover, our study stands as an exceptional example of a cross-disciplinary investigation that spanned multiple fields, including energy efficiency, data science, smart meter technology, and sustainability, to bridge the gaps between disciplines, foster collaboration, and contribute to the development of comprehensive and effective solutions. However, it is important to note that further exploration is needed to assess the reliability and practicability of our solution. Additional research and testing are necessary to validate the effectiveness and real-world applicability of our smart meter data engineering protocol and deep learning model in diverse contexts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to improve the language and check the grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

WeiQi Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zixuan Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Sybil Derrible:** Writing – review & editing, Validation. **Yangqiu Song:** Writing – review & editing, Supervision. **Zhongming Lu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The sample data and source code will be available at https://github.com/MighTy-Weaver/SMD4RAC_Detection.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.egyai.2024.100338](https://doi.org/10.1016/j.egyai.2024.100338).

References

- [1] Derribe S. *Urban engineering for sustainability*. MIT press; 2019.
- [2] Delmastro C., Abergel T., Lane K., Monschauer Y.. Cooling - International Energy Agency. 2021.
- [3] Sherman P, Lin H, McElroy M. Projected global demand for air conditioning associated with extreme heat and implications for electricity grids in poorer countries. *Energy Build* 2022;268:112198. <https://doi.org/10.1016/j.enbuild.2022.112198>.
- [4] Osunmuyiwa OO, Payne SR, Vigneswara Ilavarasan P, Peacock AD, Jenkins DP. I cannot live without air conditioning! The role of identity, values and situational factors on cooling consumption patterns in India. *Energy Res Soc Sci* 2020;69: 101634. <https://doi.org/10.1016/j.erss.2020.101634>.
- [5] Walker G, Shove E, Brown S. How does air conditioning become 'needed'? A case study of routes, rationales and dynamics. *Energy Res Soc Sci* 2014;4:1–9. <https://doi.org/10.1016/j.erss.2014.08.002>.
- [6] Zhao L, Zeng W, Yuan Z. Reduction of potential greenhouse gas emissions of room air-conditioner refrigerants: a life cycle carbon footprint analysis. *J Clean Prod* 2015;100:262–8. <https://doi.org/10.1016/j.jclepro.2015.03.063>.
- [7] Khosla R, Miranda ND, Trotter PA, Mazzone A, Renaldi R, McElroy C, et al. Cooling for sustainable development. *Nat Sustain* 2020;4:201–8. <https://doi.org/10.1038/s41893-020-00627-w>.
- [8] He P, Liu P, Qiu Y, Liu L. The weather affects air conditioner purchases to fill the energy efficiency gap. *Nat Commun* 2022;13:5772. <https://doi.org/10.1038/s41467-022-33531-2>.
- [9] Yang Z, Ding L, Xiao H, Wang B, Shi W. Field performance of household room air conditioners in Yangtze River Region in China: case studies. *J Build Eng* 2021;34: 101952. <https://doi.org/10.1016/j.jobe.2020.101952>.
- [10] Banerjee A, Solomon BD. Eco-labeling for energy efficiency and sustainability: a meta-evaluation of US programs. *Energy Policy* 2003;31:109–23. [https://doi.org/10.1016/S0301-4215\(02\)00012-5](https://doi.org/10.1016/S0301-4215(02)00012-5).
- [11] Al-Rabghi OM, Akyurt MM. A survey of energy efficient strategies for effective air conditioning. *Energy Convers Manag* 2004;45:1643–54. <https://doi.org/10.1016/j.enconman.2003.10.004>.
- [12] Asghar MR, Dan G, Miorandi D, Chlamtac I. Smart meter data privacy: a survey. *IEEE Commun Surv Tutorials* 2017;19:2820–35. <https://doi.org/10.1109/COMST.2017.2720195>.
- [13] Oh S, Haberl JS, Baltazar J-C. Analysis methods for characterizing energy saving opportunities from home automation devices using smart meter data. *Energy Build* 2020;216:109955. <https://doi.org/10.1016/j.enbuild.2020.109955>.
- [14] Carroll J, Lyons S, Denny E. Reducing household electricity demand through smart metering: the role of improved information about energy saving. *Energy Econ* 2014;45:234–43. <https://doi.org/10.1016/j.eneco.2014.07.007>.
- [15] Qi N, Cheng L, Xu H, Wu K, Li X, Wang Y, et al. Smart meter data-driven evaluation of operational demand response potential of residential air conditioning loads. *Appl Energy* 2020;279:115708. <https://doi.org/10.1016/j.apenergy.2020.115708>.
- [16] Eom J, Hyun M, Lee J, Lee H. Increase in household energy consumption due to ambient air pollution. *Nat Energy* 2020;5:976–84. <https://doi.org/10.1038/s41560-020-00698-1>.
- [17] Chen M, Sanders KT, Ban-Weiss GA. A new method utilizing smart meter data for identifying the existence of air conditioning in residential homes. *Environ Res Lett* 2019;14:094004. <https://doi.org/10.1088/1748-9326/ab35a8>.
- [18] Liu Y, Chen H, Zhang L, Wu X, Wang X. Energy consumption prediction and diagnosis of public buildings based on support vector machine learning: a case study in China. *J Clean Prod* 2020;272:122542. <https://doi.org/10.1016/j.jclepro.2020.122542>.
- [19] Zhou C, Fang Z, Xu X, Zhang X, Ding Y, Jiang X, et al. Using long short-term memory networks to predict energy consumption of air-conditioning systems. *Sustain Cities Soc* 2020;55:102000. <https://doi.org/10.1016/j.scs.2019.102000>.
- [20] Lork C, Li W-T, Qin Y, Zhou Y, Yuen C, Tushar W, et al. An uncertainty-aware deep reinforcement learning framework for residential air conditioning energy management. *Appl Energy* 2020;276:115426. <https://doi.org/10.1016/j.apenergy.2020.115426>.
- [21] Yang S, Wan MP, Chen W, Ng BF, Dubey S. Experiment study of machine-learning-based approximate model predictive control for energy-efficient building control. *Appl Energy* 2021;288:116648. <https://doi.org/10.1016/j.apenergy.2021.116648>.
- [22] Malik A, Haghdadi N, MacGill I, Ravishanker J. Appliance level data analysis of summer demand reduction potential from residential air conditioner control. *Appl Energy* 2019;235:776–85. <https://doi.org/10.1016/j.apenergy.2018.11.010>.
- [23] Wang Y, Chen Q, Hong T, Kang C. Review of smart meter data analytics: applications, methodologies, and challenges. *IEEE Trans Smart Grid* 2019;10: 3125–48. <https://doi.org/10.1109/TSG.2018.2818167>.
- [24] Liu H, Wu Y, Yan D, Hu S, Qian M. Investigation of VRF system cooling operation and performance in residential buildings based on large-scale dataset. *J Build Eng* 2022;61:105219. <https://doi.org/10.1016/j.jobe.2022.105219>.
- [25] Buzau MM, Tejedor-Aguilera J, Cruz-Romero P, Gómez-Expósito A. Detection of non-technical losses using smart meter data and supervised learning. *IEEE Trans Smart Grid* 2019;10:2661–70. <https://doi.org/10.1109/TSG.2018.2807925>.
- [26] Sanzana MR, Maul T, Wong JY, Abdulrazic MOM, Yip C-C. Application of deep learning in facility management and maintenance for heating, ventilation, and air conditioning. *Autom Constr* 2022;141:104445. <https://doi.org/10.1016/j.autcon.2022.104445>.
- [27] Hong T, Wang Z, Luo X, Zhang W. State-of-the-art on research and applications of machine learning in the building life cycle. *Energy Build* 2020;212:109831. <https://doi.org/10.1016/j.enbuild.2020.109831>.
- [28] Pereira L, Costa D, Ribeiro M. A residential labeled dataset for smart meter data analytics. *Sci Data* 2022;9:134. <https://doi.org/10.1038/s41597-022-01252-2>.
- [29] Lu Z, Broesicke OA, Chang ME, Yan J, Xu M, Derribe S, et al. Seven approaches to manage coupled human and natural systems: a sustainability toolbox. *Environ Sci Technol* 2019;53:9341–51. <https://doi.org/10.1021/acs.est.9b01982>.
- [30] T.S.I. Q-Trak. Indoor air quality monitor 7575. 2024. <https://tsi.com/products/indoor-air-quality-meters-instruments/indoor-air-quality-meters/q-trak-indoor-air-quality-monitor-7575/>.
- [31] Hochreiter S, Schmidhuber J. Long short-term memory. *Neural Comput* 1997;9: 1735–80. <https://doi.org/10.1162/neco.1997.9.8.1735>.
- [32] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need. In: *Proceedings of the 31st international conference on neural information processing systems*. Red Hook, NY, USA: Curran Associates Inc.; 2017. p. 6000–10.
- [33] Wang W, Zhou Z, Lu Z. Data-driven assessment of room air conditioner efficiency for saving energy. *J Clean Prod* 2022;338:130615. <https://doi.org/10.1016/j.jclepro.2022.130615>.
- [34] Xian Y, Lampert CH, Schiele B, Akata Z. Zero-shot learning—a comprehensive evaluation of the good, the bad and the ugly. *IEEE Trans Pattern Anal Mach Intell* 2019;41:2251–65. <https://doi.org/10.1109/TPAMI.2018.2857768>.
- [35] Kwak R-Y, Takakusagi A, Sohn J-Y, Fujii S, Park B-Y. Development of an optimal preventive maintenance model based on the reliability assessment for air-conditioning facilities in office buildings. *Build Environ* 2004;39:1141–56. <https://doi.org/10.1016/j.buildenv.2004.01.029>.
- [36] García-Sanz-Calcedo J, Gómez-Chaparro M. Quantitative analysis of the impact of maintenance management on the energy consumption of a hospital in Extremadura (Spain). *Sustain Cities Soc* 2017;30:217–22. <https://doi.org/10.1016/j.scs.2017.01.019>.
- [37] Karali N, Shah N, Park WY, Khanna N, Ding C, Lin J, et al. Improving the energy efficiency of room air conditioners in China: costs and benefits. *Appl Energy* 2020; 258:114023. <https://doi.org/10.1016/j.apenergy.2019.114023>.